Test Series No.

05

NEET Success Kit [2026-27]

DURATION: 195 Minutes | **M. MARKS: 720**

DATE: 10/06/2026

ANSWER KEY

1.	(4)	46.	(2)	91.	(2)	136.	(3)
2.	(3)	47.	(3)	92.	(4)	137.	(1)
3.	(3)	48.	(1)	93.	(1)	138.	(2)
4.	(2)	49.	(2)	94.	(2)	139.	(4)
5.	(2)	50.	(3)	95.	(1)	140.	(2)
6.	(3)	51.	(1)	96.	(3)	141.	(1)
7.	(1)	52.	(4)	97.	(2)	142.	(2)
8.	(4)	53.	(2)	98.	(2)	143.	(1)
9.	(1)	54.	(1)	99.	(3)	144.	(4)
10.	(1)	55.	(3)	100.	(4)	145.	(1)
11.	(4)	56.	(2)	101.	(3)	146.	(3)
12.	(4)	57.	(4)	102.	(4)	147.	(2)
13.	(4)	58.	(4)	103.	(2)	148.	(3)
14.	(3)	59.	(2)	104.	(3)	149.	(1)
15.	(3)	60.	(3)	105.	(1)	150.	(4)
16.	(3)	61.	(3)	106.	(4)	151.	(2)
17.	(4)	62.	(1)	107.	(2)	152.	(3)
18.	(4)	63.	(4)	108.	(4)	153.	(1)
19.	(3)	64.	(2)	109.	(3)	154.	(2)
20.	(3)	65.	(4)	110.	(2)	155.	(2)
21.	(1)	66.	(4)	111.	(4)	156.	(2)
22.	(1)	67.	(2)	112.	(2)	157.	(4)
23.	(2)	68.	(4)	113.	(4)	158.	(3)
24.	(3)	69.	(2)	114.	(2)	159.	(2)
25.	(4)	70.	(2)	115.	(2)	160.	(4)
26.	(2)	71.	(4)	116.	(1)	161.	(3)
27.	(2)	72.	(4)	117.	(4)	162.	(3)
28.	(3)	73.	(4)	118.	(3)	163.	(3)
29.	(2)	74.	(4)	119.	(2)	164.	(3)
30.	(4)	75.	(1)	120.	(3)	165.	(3)
31.	(4)	76.	(1)	121.	(1)	166.	(4)
32.	(3)	77.	(4)	122.	(2)	167.	(4)
33.	(1)	78.	(3)	123.	(4)	168.	(2)
34.	(3)	79.	(3)	124.	(4)	169.	(2)
35.	(2)	80.	(2)	125.	(2)	170.	(2)
36.	(1)	81.	(2)	126.	(3)	171.	(4)
37.	(3)	82.	(1)	127.	(4)	172.	(4)
38.	(2)	83.	(3)	128.	(2)	173.	(3)
39.	(2)	84.	(4)	129.	(4)	174.	(3)
40.	(3)	85.	(2)	130.	(2)	175.	(3)
41.	(1)	86.	(2)	131.	(3)	176.	(3)
42.	(1)	87.	(1)	132.	(4)	177.	(4)
43.	(2)	88.	(1)	133.	(1)	178.	(3)
44.	(3)	89.	(1)	134.	(2)	179.	(2)
45.	(1)	90.	(1)	135.	(3)	180.	(1)

[Contd...

1. (4)

$$E = \frac{hc}{\lambda}$$

$$\frac{\lambda}{hc} = \frac{1}{E} \Rightarrow \frac{\lambda^2}{hc} = \frac{\lambda}{E}$$

$$\left[\frac{\lambda^2}{hc}\right] = \left[\frac{\lambda}{E}\right] = \left[\frac{L}{ML^2T^{-2}}\right] = \left[M^{-1}L^{-1}T^2\right]$$

(NCERT Ed. 2025-26, 11th, Page No. 11)

2. (3)

For stable equilibrium, $\theta = 0°$

$U = -MB$

$= -0.32 \times 0.15 = -4.8 \times 10^{-2}$ J

(NCERT Ed. 2025-26, 12th, Page No. 139)

3. (3)

gauge pressure = 8 atm – 1 atm

= 7 atm = 7×10^5 pascal

(NCERT Ed. 2025-26, 11th, Page No. 183)

4. (2)

$$R_P = \frac{1200 \times 400}{1200 + 400} = 300\,\Omega$$

Reading of voltmeter

$$= \frac{300}{(300 + 600)} \times 300 = 100\,\text{V}$$

(NCERT Ed. 2025-26, 12th, Page No. 102)

5. (2)

In the direction of electric field, electric potential decreases. Here $V_A = V_B = V_C$

(NCERT Ed. 2025-26, 12th, Page No. 55)

6. (3)

$$a = \frac{F}{m} = \frac{F}{13}\text{m/s}^2$$

$v = u + at$

$$v = 0 + \frac{F}{13}26$$

$v = 2F$ m/s

For next 15 s

$s = vt$

$75 = 2F \times 15$

$F = 2.5$ N

(NCERT Ed. 2025-26, 11th, Page No. 54)

7. (1)

$$R_d = \frac{1.1 - 0.8}{(35 - 20) \times 10^{-3}} = 20\Omega$$

(NCERT Ed. 2025-26, 12th, Page No. 337)

8. (4)

According to Wein's displacement law,

$$\lambda = \frac{b}{T}$$

i.e.. $\lambda \propto \frac{1}{T}$ …(i)

We know that,

$\lambda_{\text{bluish}} < \lambda_{\text{yellowish}} < \lambda_{\text{reddish}}$

Hence, using E.q. (i), we have

$T_A > T_C > T_B$

(NCERT Ed. 2025-26, 11th, Page No. 218)

9. (1)

In figure A, each material behave as separate convex lens, so 4 image are formed.

In figure B, there in one effective focal length of the combination, so 1 image will formed.

(NCERT Ed. 2025-26, 12th, Page No. 238)

10. (1)

$$I_P = I_0 + M\left(\frac{3R}{2}\right)^2$$

$$= \frac{MR^2}{2} + \frac{9MR^2}{4}$$

$$I_P = \frac{11MR^2}{4} = MK^2$$

$$K = \sqrt{\frac{11}{4}}R$$

(NCERT Ed. 2025-26, 11th, Page No. 115)

11. (4)

$$|e| = \frac{BdA}{dt}$$

$$= B \times 2\pi r \times \frac{dr}{dt}$$

$= 0.025 \times 2\pi \times 2 \times 10^{-2} \times 3 \times 10^{-3}$

$= 3\ \pi\mu$V

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[Contd...

12. (4)

Total energy, $E = \frac{1}{2}ka^2$

$E = \frac{1}{2}m\omega^2 a^2 \Rightarrow E \propto a^2$

$\frac{E'}{E} = \frac{\left(\frac{3}{4}a\right)^2}{a^2} \Rightarrow E' = \frac{9}{16}E$

(NCERT Ed. 2025-26, 11th, Page No. 269)

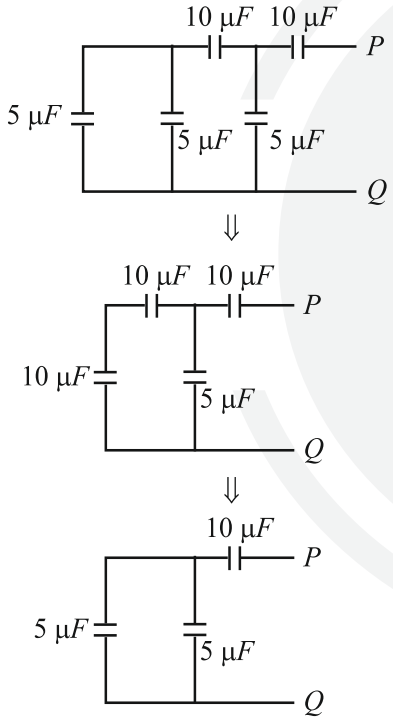
13. (4)

Drunkard walks 2m forward in 8 second. So, he cover 8 m forward in 32 second. After 8 m, he moves 5 steps forward in 5 s and reach to pit and fall into it.

Total time = 32 + 5 = 37 s

(NCERT Ed. 2025-26, 11th, Page No. 24)

14. (3)



$C_{eq} = 5\ \mu F$

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15. (3)

A. For diffraction pattern, the size of slit should be comparable to the wavelength of wave used.

B. Diffraction phenomenon is commonly observed in our daily routine in case of sound waves (which is a longitudinal wave) because wavelength of sound waves is large.

C. Diffraction is a wave phenomenon. It is observed in electromagnetic waves and longitudinal waves as well.

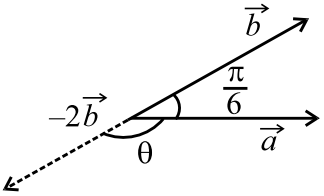
(NCERT Ed. 2025-26, 12th, Page No. 266)

16. (3)

The speed of satellite changes with position and its potential energy also changes. The total energy remains constant, because only conservative force (gravitational force) is acting.

(NCERT Ed. 2025-26, 11th, Page No. 63)

17. (4)



$\theta = \pi - \frac{\pi}{6}$

$= \frac{5\pi}{6}$

(NCERT Ed. 2025-26, 11th, Page No. 29)

18. (4)

$Q = 2(4 \times 7.06) - (7 \times 5.60)$

$Q = 17.3$ MeV

(NCERT Ed. 2025-26, 12th, Page No. 312)

19. (3)

$T = \frac{PV}{nR}$

$T_A = \frac{\left(5\times10^4\right)\times10}{2R}$

$T_B = \frac{\left(10\times10^4\right)\times10}{2R}$

$T_C = \frac{\left(10\times10^4\right)\times20}{2R}$

$T_A : T_B : T_C = 1:2:4$

(NCERT Ed. 2025-26, 11th, Page No. 204)

20. (3)

$\vec{\tau} = i\left(\vec{A}\times\vec{B}\right)$

$\vec{\tau} = 5\times\left[\left(4\hat{j}\right)\times\left(2\hat{i}+2\hat{j}+4\hat{k}\right)\right]$

$= 5\times\left[16\hat{i}-8\hat{k}\right]$

$|\vec{\tau}| = 5\sqrt{(16)^2+(8)^2}$

$= 5\sqrt{320}\ \text{N}-\text{m}$

(NCERT Ed. 2025-26, 12th, Page No. 125)

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21. (1)

For point charge, $E \propto \frac{1}{r^2}$

For line charge, $E \propto \frac{1}{r}$

For infinite plane sheet, $E \propto r^0$

For short electric dipole, $E \propto \frac{1}{r^3}$

(NCERT Ed. 2025-26, 12th, Page No. 40)

22. (1)

In 6th overtone,

No of nodes = 8

No of antinodes = 7

(NCERT Ed. 2025-26, 11th, Page No. 291)

23. (2)

$$i = \frac{20 - 0.7 - 0.3}{R_{eq}} = \frac{19}{5}\text{mA}$$

$i = 3.8$ mA

$V = 3.8 \times 10^{-3} \times 2 \times 10^3$

$= 7.6$ V

(NCERT Ed. 2025-26, 12th, Page No. 335)

24. (3)

$$e = \frac{v_2 - v_1}{u_1 - u_2}$$

For heavy wall, $u_2 = v_2$

$1 = \frac{10 - v_1}{-30 - 10}$ ($e = 1$, for elastic collision)

$-40 = 10 - v_1$

$v_1 = 50$ m/s

(NCERT Ed. 2025-26, 11th, Page No. 84)

25. (4)

$P_i = V_i \times I_i$

$5000 = 200 \times I_i$

$I_i = 25\text{A}$

$\eta = \frac{P_0}{P_i}$

$P_0 = 0.70 \times 5000$

$P_0 = 3500$ watt

$3500 = V_0 \times I_0$

$V_0 = \frac{3500}{2} = 1750$ volt

(NCERT Ed. 2025-26, 12th, Page No. 195)

26. (2)

Plane angle is dimensionless variable, π is dimensionless constant.

(NCERT Ed. 2025-26, 11th, Page No. 10)

27. (2)

$\frac{X}{60} = \frac{10}{40} \Rightarrow 15\Omega$

(NCERT Ed. 2025-26, 12th, Page No. 101)

28. (3)

Interference occurs due to superposition of light waves from two coherent source. If one slit is closed, interference not occur.

(NCERT Ed. 2025-26, 12th, Page No. 265)

29. (2)

$T_A = 30^\circ\text{C}$

$T_B = 90^\circ\text{F} \approx 32^\circ\text{C}$

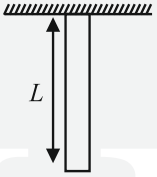
$T_C = 200\text{K} = -73^\circ\text{C}$

$T_D = 60^\circ\text{C}$

(NCERT Ed. 2025-26, 11th, Page No. 204)

30. (4)

Maximum stress is at the top.



$\frac{F}{A} \le \sigma_0$

$\frac{mg}{A} \le \sigma$

$\frac{d \times (AL) \times g}{A} \le \sigma$

$L \le \frac{\sigma}{dg}$

$L_{max} = \frac{\sigma}{dg}$

(NCERT Ed. 2025-26, 11th, Page No. 168)

31. (4)

Power factor = cos ϕ

Cos ϕ cannot be greater than 1

(NCERT Ed. 2025-26, 12th, Page No. 191)

32. (3)

$r = \frac{mv}{qB} = \frac{p}{qB} = \frac{\sqrt{2mE}}{qB}$

q, E and B are same for both particle

$\frac{r_1}{r_2} = \frac{\sqrt{m_1}}{\sqrt{m_2}}$

$\frac{m_1}{m_2} = \left(\frac{r_1}{r_2}\right)^2$

(NCERT Ed. 2025-26, 12th, Page No. 112)

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33. (1)

$\text{Thrust} = -u\frac{dm}{dt}$

$\text{Thrust} = 10000 \times 16 = 1.6 \times 10^5$ N

(NCERT Ed. 2025-26, 11th, Page No. 54)

34. (3)

$$\mu = \frac{\sin\left(\frac{A+\delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

$$\mu = \frac{\sin\left(\frac{60°}{2}+\frac{\delta_m}{2}\right)}{\sin\left(\frac{60°}{2}\right)} = \frac{\sin\left(30°+\frac{\delta_m}{2}\right)}{\frac{1}{2}}$$

$$= 2\sin\left(30°+\frac{\delta_m}{2}\right)$$

(NCERT Ed. 2025-26, 12th, Page No. 240)

35. (2)

$E_0 = CB_0$

$= 3 \times 10^8 \times 20 \times 10^{-9} = 6$ V/m

(NCERT Ed. 2025-26, 12th, Page No. 207)

36. (1)

$$V_{rms} = \sqrt{\frac{3RT}{M}}$$

$$1932 = \sqrt{\frac{3\times 8.3\times 300}{M}}$$

$$M = \frac{3\times 8.3\times 300}{(1932)^2} = 0.002 \text{ kg}$$

(NCERT Ed. 2025-26, 11th, Page No. 250)

37. (3)

Lyman series → electron jump for higher energy level to $n = 1$

Balmer series → electron jump from higher energy level to $n = 2$

Paschen series → electron jump from higher energy level to $n = 3$

Brackett series → electron jump from higher energy level to $n = 4$

(NCERT Ed. 2025-26, 12th, Page No. 300)

38. (2)

3rd overtone = 4(fundamental frequency)

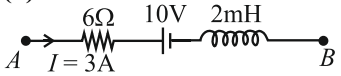
Third overtone is 4th harmonic

$4f = 500$

$f = 125$ Hz

(NCERT Ed. 2025-26, 11th, Page No. 291)

39. (2)



$V_A - (3\times 6) - 10 - (2\times 10^{-3}\times 10^3) = V_B$

$\Rightarrow V_A - V_B = 30$ volt

(NCERT Ed. 2025-26, 12th, Page No. 168)

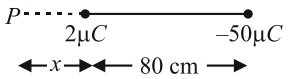
40. (3)

If magnitude of average velocity and average speed is same for the time interval, then particle must move in straight line without turning back.

Equation of kinematics is used only for object having constant acceleration.

(NCERT Ed. 2025-26, 11th, Page No. 39)

41. (1)



$E_{\text{at } p} = 0$

$$\frac{kq_1}{x^2} - \frac{kq_2}{(80+x)^2} = 0$$

$$\frac{q_1}{x^2} = \frac{q_2}{(80+x)^2}$$

$$\frac{2}{(x)^2} = \frac{50}{(80+x)^2}$$

$$\frac{80+x}{x} = 5$$

$x = 20$ cm.

(NCERT Ed. 2025-26, 12th, Page No. 12)

42. (1)

$$L = \frac{nh}{2\pi} \Rightarrow L \propto n$$

$$\frac{L_2}{L_1} = \frac{2}{1}$$

(NCERT Ed. 2025-26, 12th, Page No. 299)

43. (2)

$$\frac{1}{-36} = (1.5-1)\left(\frac{1}{3R} - \frac{1}{R}\right)$$

$R = 12$ cm

$\therefore\ 3R = 36$ cm

(NCERT Ed. 2025-26, 12th, Page No. 234)

44. (3)

Volume flow rate = Area × velocity of liquid through orifice

$$\frac{dV}{dt} = A\times\sqrt{2gh}$$

(NCERT Ed. 2025-26, 11th, Page No. 189)

45. (1)

$P = \vec{F}\cdot\vec{V}$

$= \left(2\hat{i}+\hat{j}-3\hat{k}\right).\left(2\hat{i}-3\hat{j}\right) = 4 - 3 = 1$ watt

(NCERT Ed. 2025-26, 11th, Page No. 83)

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46. (2)

- Positive deviation happens when A–B interactions are weaker than A–A and B–B → molecules escape more easily → vapour pressure increases.
- Ethanol + water: mixing disrupts H-bond network; A–B interactions become effectively weaker than pure components → positive deviation.
- Acetone + chloroform forms strong H-bonding → negative deviation.
- Hexane + heptane and benzene + toluene are similar → nearly ideal.

(NCERT Ed. 2025-26, 12th, Page No. 14)

47. (3)

Molten $AlCl_3$ gives:

$Al^{3+} + 3e^- \rightarrow Al$

So 3 moles of electrons per mole Al.

$13.5\text{ g Al} = \frac{13.5}{27} = 0.5\text{ mol Al}$

Total Faraday = 0.5 × 3 = 1.5 F

(NCERT Ed. 2025-26, 12th, Page No. 56)

48. (1)

Arrhenius:

$$\ln k = \ln A - \frac{E_a}{R}\left(\frac{1}{T}\right)$$

$$\text{So slope} = -\frac{E_a}{R}$$

$$\text{Given slope} = -9.0\times10^3$$

$$\frac{E_a}{R} = 9.0\times10^3$$

$$E_a = 9.0\times10^3 \times 8.314$$

$$E_a = 74826\text{ J mol}^{-1} = 74.8\text{ kJ mol}^{-1}$$

(NCERT Ed. 2025-26, 12th, Page No. 80)

49. (2)

Co(III) = +3; NH_3 neutral; Cl^- inside = -1

Complex charge = +3 – 1 = +2 → needs 2 Cl^- outside.

(NCERT Ed. 2025-26, 12th, Page No. 80)

50. (3)

Primary amines ($R–NH_2$) react with benzenesulphonyl chloride to form sulphonamides containing an acidic N–H hydrogen.

Due to the strong electron-withdrawing $–SO_2–$ group, this hydrogen is acidic and the sulphonamide dissolves in NaOH forming a soluble salt.

(NCERT Ed. 2025-26, 12th, Page No. 271)

51. (1)

Vitamin A	Xerophthalmia
Vitamin B_1	Beri-beri
Vitamin C	Scurvy
Vitamin D	Rickets

(NCERT Ed. 2025-26, 12th, Page No. 296)

52. (4)

K_f depends only on the solvent, not on molality or solute amount.

(NCERT Ed. 2025-26, 12th, Page No. 18)

53. (2)

Relations: $\kappa = G \times (\text{cell constant})$

and $G = \frac{1}{R}$

So, $\text{cell constant} = \kappa \times R$

Given: $\kappa = 0.015\text{ S cm}^{-1}$

$R = 40\ \Omega$

$$\text{cell constant} = 0.015\times40 = 0.60\text{ cm}^{-1}$$

(NCERT Ed. 2025-26, 12th, Page No. 41)

54. (1)

Basicity of group 15 hydrides decreases down the group.

55. (3)

Moles $\frac{4}{2} = 2\text{ mol}$.

Molecules = 2 × 6.022 × 10^{23} = 1.2044 × 10^{24}.

(NCERT Ed. 2025-26, 11th, Page No. 18)

56. (2)

$$\text{Moles of urea} = \frac{6}{60} = 0.1\text{ mol}$$

$$\text{Molality} = \frac{0.1}{0.1} = 1\text{ mol/kg}$$

$$\Delta T_b = 0.52\times1 = 0.52\text{ K}$$

(NCERT Ed. 2025-26, 12th, Page No. 17)

57. (4)

$\Delta G° = –nFE°_{cell}$

$\Delta G° = –2 \times 96500 \times 1.10 = 212300\text{ J mol}^{-1}$

So, $\Delta G° = –212.30\text{ kJ mol}^{-1}$

(NCERT Ed. 2025-26, 12th, Page No. 40)

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58. (4)

$$\underset{\text{Methyl bromide}}{CH_3Br} \xrightarrow[-KBr]{KCN} \underset{\substack{\text{Ethane nitrile}\\ A}}{CH_3CN} \xrightarrow[\text{Complete hydrolysis}]{H_3O^+} \underset{\substack{\text{Ethanoic acid}\\ B}}{CH_3COOH}$$

$$\xrightarrow[\text{Ether}]{LiAlH_4} \underset{\substack{\text{Ethanol}\\ C}}{CH_3CH_2OH}$$

(NCERT Ed. 2025-26, 12th, Page No. 170)

59. (2)

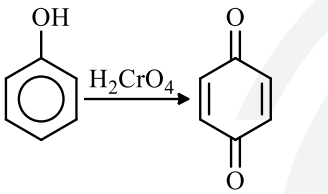
Let $p_{CO_2} = x$, $p_{CO} = 1 - x$. Solids are omitted.

$$K_p = \frac{p_{CO_2}}{p_{CO}} = \frac{x}{1-x} = 0.5$$

$x = 0.5 - 0.5\,x, 1.5\,x = 0.5, x = 0.333$ atm

(NCERT Ed. 2025-26, 11th, Page No. 183)

60. (3)



(Benzoquinone)

(NCERT Ed. 2025-26, 12th, Page No. 213)

61. (3)

Basic amino acids have an **extra basic group** in the side chain and can accept H^+.

(NCERT Ed. 2025-26, 12th, Page No. 268)

62. (1)

A combination redox reaction involves elements combining, with oxidation and reduction occurring. $C + O_2 \rightarrow CO_2$ has C (0 to +4) oxidized and O (0 to –2) reduced.

(NCERT Ed. 2025-26, 11th, Page No. 242)

63. (4)

Thiosulphate ($S_2O_3^{2-}$) reacts with dilute acids (like HCl) to form colloidal sulphur, giving a white turbidity/precipitate:

$S_2O_3^{2-} + 2H^+ \rightarrow SO_2 + S\downarrow + H_2O$

Passing excess H_2S can produce more sulphur, giving yellow sulphur precipitate.

Hence, the anion is thiosulphate.

64. (2)

Electron withdrawing groups like $–NO_2$, –CN, –CHO, –COOH, etc, when present at the ortho and para-positions with respect to phenolic group increases the acidity of phenol due to greater stabilisation of phenoxide ion. While the presence of electron releasing group like $–NH_2$, $–CH_3$, etc, decrease the acidity of phenols. This explains the following order of acidity

p-nitrophenol > phenol > p-cresol.

(NCERT Ed. 2025-26, 12th, Page No. 207)

65. (4)

$Al_2(SO_4)_3 \rightarrow 2Al^{3+} + 3SO_4^{2-} \Rightarrow i = 5$;

and $K_4\left[Fe(CN)_6\right] \rightarrow 4K^+\left[Fe(CN)_6\right]^{4-} \Rightarrow i = 5$

(NCERT Ed. 2025-26, 12th, Page No. 22)

66. (4)

- NO → Neutral
- SO_3 → Acidic
- Bi_2O_3 → Basic
- As_2O_3 → Amphoteric

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67. (2)

$r = k[A]^m[B]^n$

Doubling $[A]$ makes rate 4 times:

$2^m = 4 \Rightarrow m = 2$

Doubling $[B]$ makes rate 2 times:

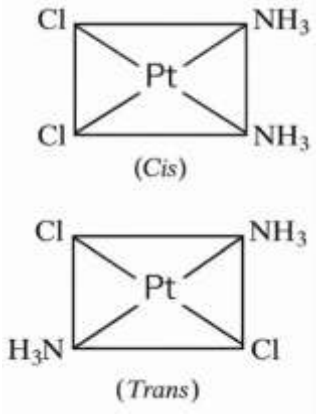
$2^n = 2 \Rightarrow n = 1$

So, $r = k[A]^2[B]$

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68. (4)

Geometrical isomers for $[Pt(NH_3)_2Cl_2]$ is



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69. (2)

All halogens are coloured. This is due to absorption of radiations in visible region which results in the excitation of outer electrons to higher energy level.

[Contd...

70. (2)

Lactose is a disaccharide made of galactose + glucose.

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71. (4)

Carbylamine test is responded by primary or 1°-amine (aliphatic or aromatic) only.

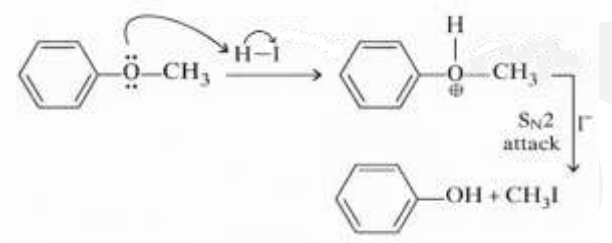
$$R\text{–}NH_2 \xrightarrow[KOH/\Delta]{CHCl_3} R\text{–}NC + KCl + H_2O$$

$$Ar\text{–}NH_2 \xrightarrow[KOH/\Delta]{CHCl_3} Ar\text{–}NC + KCl + H_2O$$

Carbylamine

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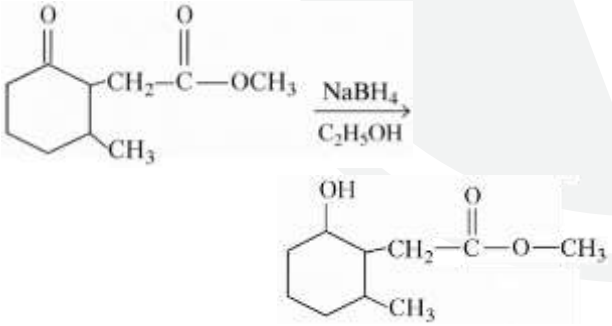
72. (4)



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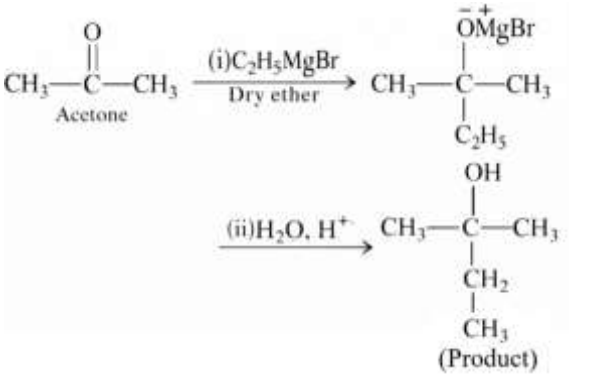
73. (4)

$NaBH_4$ is a weak reducing agent. It can reduce aldehyde/ketone to alcohol but cannot reduce ester group.



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74. (4)



IUPAC name of product is 2-methylbutan-2-ol.

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75. (1)

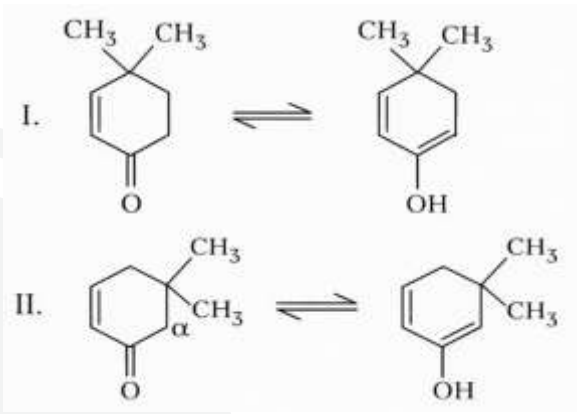
Check $i \times m$:

- NaCl: $2 \times 0.010 = 0.020$
- $MgCl_2$: $3 \times 0.006 = 0.018$
- glucose: $1 \times 0.012 = 0.012$
- Na_2SO_4: $3 \times 0.004 = 0.012$

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76. (1)

In keto–enol tautomerism keto form should have α-hydrogen (structure I and II).



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77. (4)

For C_2: 12 electrons,

$(\sigma 1s)^2(\sigma^* 1s)^2(\sigma 2s)^2(\sigma^* 2s)^2(\pi 2p_x)^2(\pi 2p_y)^2$

Bonding electron = 8, antibonding electron = 4,

Bond order $= \frac{1}{2}(8-4) = 2$

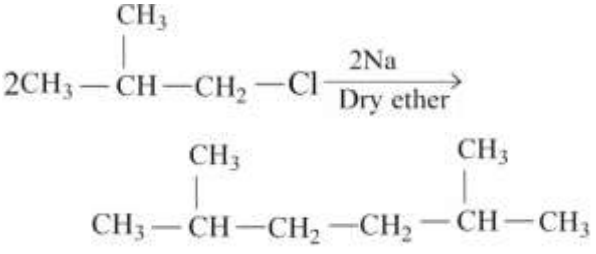
(NCERT Ed. 2025-26, 11th, Page No. 130)

78. (3)

$^{64}Gd = [Xe]\ 4f^7\ 5d^1\ 6s^2$

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79. (3)



(NCERT Ed. 2025-26, 12th, Page No. 181)

80. (2)

Work done on the system (compression, $\Delta V < 0$) is positive ($w = -p\Delta V > 0$).

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[Contd...

81. (2)

$XeF_6 + 3H_2O \rightarrow XeO_3 + 6HF$

Noble gases have low boiling points because they have weak London Dispersion Forces (van der Waals forces), not strong hydrogen bonding.

82. (1)

- 4-methylphenol = p-cresol → A–II
- benzene-1,2-diol = catechol → B–III
- benzene-1,3-diol = resorcinol → C–I
- benzene-1,4-diol = hydroquinone → D–IV

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83. (3)

The Gabriel synthesis is a method used for the preparation of primary amines via an S_N2 (nucleophilic substitution) reaction between a phthalimide anion and an alkyl halide. For the S_N2 reaction to proceed with a good yield, the alkyl halide must be unhindered.

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84. (4)

$$K_a = \frac{[H^+][F^-]}{[HF]} \approx \frac{x^2}{0.1}, 6.8\times10^{-4} = \frac{x^2}{0.1}$$

$$x^2 = 6.8\times10^{-5}, x \approx 8.25\times10^{-3}$$

$$pH = -\log(8.25\times10^{-3}) \approx 2.08$$

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85. (2)

One-eighth means: $\frac{[A]}{[A]_0} = \frac{1}{8} = \left(\frac{1}{2}\right)^3$

So it takes 3 half-lives.

First-order half-life: $t_{1/2} = \frac{0.693}{k}$

$$= \frac{0.693}{2.31\times10^{-3}} = 300\,s$$

Total time: $t = 3t_{1/2} = 3\times300 = 900$ s

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86. (2)

The CFSE for octahedral complex is given by

CFSE = $[-0.4\ t_{2g}e^- + 0.6\ e_ge^-]$

For Fe^{3+}, $[3d^5] \rightarrow t_{2g}^3\ e_g^2$

CFSE = $[(-0.4\times3) + (0.6\times2)] = 0$

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87. (1)

S_N1 reaction is carried out in two steps. In form of slow step

Step (i) carbonium (carbocation) ion is formed and its formation is based upon the stability

Stability order of carbocation benzylic carbocation (resonating stable)

$$C_6H_5\overset{+}{C}H_2 > CH_3-\underset{2^\circ\text{ carbocation}}{\overset{+}{C}H}-CH_3 > CH_3-\underset{1^\circ\text{ carbocation}}{\overset{+}{C}H_2}$$

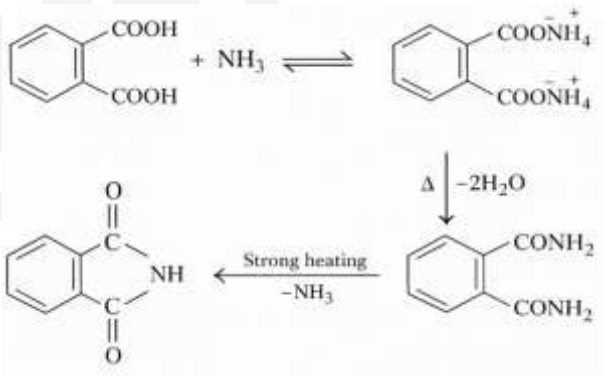
(NCERT Ed. 2025-26, 12th, Page No. 174)

88. (1)

$$CH_3CH_2Br + KCN \xrightarrow{\text{alcohol}} CH_3CH_2CN + KBr$$

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89. (1)



(NCERT Ed. 2025-26, 12th, Page No. 253)

90. (1)

Find d-electrons (3d series, remove 4s first):

- Ti = [Ar] $3d^24s^2 \rightarrow Ti^{3+} = 3d^1 \rightarrow n = 1$
- V = $3d^34s^2 \rightarrow V^{2+} = 3d^3 \rightarrow n = 3$
- Mn = $3d^54s^2 \rightarrow Mn^{2+} = 3d^5 \rightarrow n = 5$
- Zn = $3d^{10}4s^2 \rightarrow Zn^{2+} = 3d^{10} \rightarrow n = 0$

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[Contd...

91. (2)

Ethidium bromide binds to DNA and fluoresces under UV light, allowing the visualization of DNA fragments as bright orange bands. Pure DNA fragments cannot be seen in visible light without staining.

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92. (4)

The site of glycolytic pathway in case of organism undergoing aerobic and anaerobic respiration in cytoplasm.

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93. (1)

- Golgi apparatus performs the function of packaging of material.
- In *Amoeba* the contractile vacuole is important for osmoregulation and excretion.
- The nuclear matrix or the nucleoplasm contains nucleolus and chromatin.

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94. (2)

Sodium-potassium pumps move across cell membrane by active transport.

In human beings, the membrane of the erythrocyte has approximately 52 per cent protein and 40 per cent lipids.

Active transport is energy dependent process.

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95. (1)

Adaptive radiation is the process by which a single ancestral species evolves into multiple species, each adapted to a specific ecological niche. This phenomenon is often observed in isolated environments, such as islands, where there are fewer competitors and a variety of unexploited resources. Darwin's finches and Australian marsupials are classic examples of adaptive radiation, but it is not limited to these groups. Geographical barriers are not a requirement for adaptive radiation, and it does not result in identical species but rather in diverse forms adapted to different niches.

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96. (3)

List-I	List-II
Denaturation	Separation of DNA strands by heating at ~94°C
Annealing	Attachment of primers to complementary sequences at template DNA
Primers	Chemically synthesized oligonucleotides complementary to DNA regions
Thermostable DNA polymerase	Used in extension of primers

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97. (2)

RuBisCO (Ribulose-1, 5-bisphosphate carboxylase /oxygenase) is present in both C_3 and C_4 plants.

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98. (2)

In the rocky intertidal communities of the American Pacific Coast the starfish *Pisaster* is an important predator. In a field experiment, when all the starfish were removed from an enclosed intertidal area, more than 10 species of invertebrates became extinct within a year, because of inter-specific competition.

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99. (3)

Phycomycetes - Coenocytic

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100. (4)

- In a terrestrial ecosystem, a much larger fraction of energy flows through the detritus food chain than through the GFC.
- Detritus food chain may be connected with the grazing food chain at some levels: some of the organisms of DFC are prey to the GFC animals, and in a natural ecosystem, some animals like cockroaches, crows, etc., are omnivores.

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101. (3)

Sponges exhibit indirect development with a larval stage that is morphologically distinct from the adult. Sponges have a cellular level of organization, a water transport system, and intracellular digestion. Sponges reproduce asexually by fragmentation and sexually by formation of gametes.

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[Contd...

102. (4)

- Glandular epithelium is correctly described as including goblet cells of the alimentary canal as an example of unicellular glands.
- Exocrine glands secrete products like mucus and saliva through ducts, not hormones.
- Endocrine glands secrete hormones directly into the bloodstream, not through ducts.
- Cuboidal epithelium is found in ducts of glands and tubular parts of nephrons, not in the lining of the stomach and intestine.

(NCERT Ed. 2020-21, 11th, Page No. 102)

103. (2)

- Bacteriophages have a high copy number of their genome within bacterial cells, making them suitable for cloning large amounts of foreign DNA.
- Both bacteriophages and plasmids can replicate independently of chromosomal DNA, and both can be used as vectors for cloning. Plasmids are not the only vectors used in cloning.

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104. (3)

XO male	Grasshopper
XY male	*Drosophila*
Haploid male	Honey bee
ZZ male	Bird

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105. (1)

Insulin is a peptide hormone that gets broken down by proteolytic enzymes (like pepsin and trypsin) in the digestive system, making oral administration ineffective. This is why insulin is usually given via subcutaneous injection to bypass digestion.

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106. (4)

The *vasa recta* is formed by the efferent arteriole, not the afferent arteriole. It is a U-shaped capillary network that runs parallel to the loop of Henle and is highly developed in juxtamedullary nephrons, playing a key role in maintaining the medullary osmotic gradient.

The vasa recta is not involved in filtration; filtration occurs in the glomerulus. It is highly developed in juxtamedullary nephrons, not cortical nephrons.

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107. (2)

Dense irregular connective tissue has collagen fibers oriented in multiple directions, providing strength in different directions (e.g., dermis of the skin). Dense regular connective tissue has parallel fibers, found in tendons and ligaments. Areolar and adipose tissues do not have dense collagen fibers.

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108. (4)

The Z-line is located in the center of the I-band, not the A-band. It bisects the I-band and anchors the thin (actin) filaments, not the thick (myosin) filaments. The A-band contains both actin and myosin filaments and is bisected by the M-line, which holds the thick filaments together. The H-zone is the central region of the A-band where only myosin filaments are present, as it is not overlapped by actin filaments.

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109. (3)

The transgenic plant in (b) is genetically engineered to produce double-stranded RNA (dsRNA), which triggers RNA interference (RNAi) in the nematode. This silences critical nematode genes necessary for their survival and development, thereby preventing infestation.

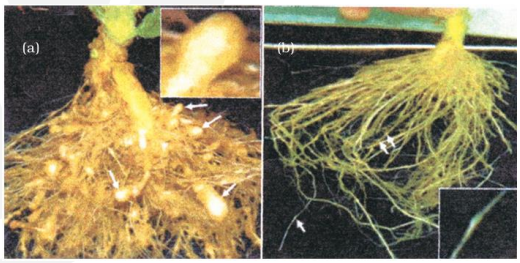


Fig: Host plant-generated dsRNA triggers protection against nematode infestation:
(a) Roots of a typical control plants: (b) transgenic plant roots 5 days after deliberate infection of nematode but protected through novel mechanism.

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110. (2)

The Hardy-Weinberg equation is represented as $p^2 + 2pq + q^2 = 1$, where p and q are allele frequencies.

Hardy-Weinberg equilibrium applies only to ideal populations where no evolutionary forces (such as natural selection, gene flow, genetic drift, mutations, or recombination) are acting. In real populations, these factors usually cause deviations from equilibrium, leading to evolutionary change.

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[Contd...

111. (4)

List-I	List-II
Purkinje fibres	Conduct electrical impulses to the ventricular musculature
Pericardium	Protective membrane surrounding the heart
Chordae tendinae	Anchor the AV valve cusps to prevent backflow of blood
Bundle of His	Conducts electrical impulses from the AV node to the ventricles

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112. (2)

The secondary oocyte formed during the first meiotic division retains most of the nutrient-rich cytoplasm of the primary oocyte.
The Graafian follicle releases the secondary oocyte (ovum) during ovulation, which is surrounded by a new membrane called the zona pellucida.

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113. (4)

Axial skeleton comprises 80 bones distributed along the main axis of the body. The skull, vertebral column, sternum and ribs constitute axial skeleton.
The vertebral column is differentiated into cervical (7), thoracic (12), lumbar (5), sacral (1-fused) and coccygeal (1-fused) regions starting from the skull.

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114. (2)

Dinosaurs evolved from the thecodonts, which were a group of archosaurian reptiles present during the Triassic period. Thecodonts gave rise to both dinosaurs and modern crocodiles. Pelycosaurs and synapsids are related to mammals rather than dinosaurs, while therapsids are ancestors of mammals.

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115. (2)

The primary consumers are herbivores.

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116. (1)

The sporophyte of liverworts is differentiated into a foot, seta and capsule. After meiosis, spores are produced within the capsule. These spores germinate to form free-living gametophytes.

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117. (4)

Carboxylation is the most crucial step of the Calvin cycle where CO_2 is utilised for the carboxylation of RuBP.

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118. (3)

In cockroaches, each thoracic segment bears a pair of walking legs. Males bear a pair of short, thread like anal styles which are absent in females. Hindwings are transparent and used in flight.

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119. (2)

The correct sequence which shows the development of females gametophyte in flowering plants is:
Megaspore mother cell $\xrightarrow{\text{meiosis}}$ megaspore $\xrightarrow{\text{mitosis}}$ female gametophyte.

(NCERT Ed. 2025-26, 12th, Page No. 9)

120. (3)

A sigmoid curve is a characteristic of living organism growing in a natural environment.
Root elongating at a constant rate shows arithmetic growth.

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121. (1)

List-I	List-II
Body covering	Chitinous exoskeleton
Respiratory organs	Gills, book gills, book lungs, or tracheal system
Excretory organs	Malpighian tubules
Circulatory system	Open type

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122. (2)

- The given diagram represents *Mucor,* member of phycomycetes.
- The mycelium is aseptate and coenocytic. Asexual reproduction takes place by zoospores (motile) or by aplanospores (non-motile). These spores are endogenously produced in sporangium.

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123. (4)

Branch of taxonomy which is based on cytological information like chromosome number, structure and behaviour is called as cytotaxonomy.

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124. (4)

List-I	List-II
Tidal Volume (TV)	500 mL
Inspiratory Reserve Volume (IRV)	2500 mL – 3000 mL
Expiratory Reserve Volume (ERV)	1000 mL – 1100 mL
Residual Volume (RV)	1100 mL – 1200 mL

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125. (2)

Essential amino acids are those that cannot be synthesized by the human body in sufficient amounts and must be obtained from dietary sources. These amino acids are crucial for protein synthesis, enzyme function, and overall metabolism.

Essential amino acids are crucial for protein synthesis because proteins are made up of amino acids, and without essential ones, the process would be incomplete.

Essential amino acids are found in both plant and animal sources.

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126. (3)

Erythroxylum coca produces cocaine, a stimulant with potential hallucinogenic effects at high doses.

Atropa belladonna contains tropane alkaloids like atropine and scopolamine, which can cause hallucinations. *Datura* contains scopolamine and atropine, which are known to cause hallucinations.

Papaver somniferum produces opium, which contains morphine and codeine. These are pain-relieving drugs and do not have hallucinogenic effects.

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127. (4)

If the age distribution (per cent individuals of a given age or age group) is plotted for the population, the resulting structure is called an age pyramid.

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128. (2)

O_2 evolved by the green plant comes from H_2O, not from carbon dioxide. This was proved by using radioisotopic techniques.

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129. (4)

Base pairs in each turn of DNA helix (roughly)	10
Hydrogen bonds between cytosine and guanine	3
Length of DNA in humans in metres	2.2
Length of DNA in *E.coli* in mm	1.36

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130. (2)

Aspergillus niger	Citric acid
Acetobacter aceti	Acetic acid
Clostridium butylicum	Butyric acid
Lactobacillus	Lactic acid

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131. (3)

Oxytocin, released from the maternal pituitary, acts on the uterine muscles to induce stronger contractions. This creates a positive feedback loop, where stronger contractions stimulate more oxytocin release, ultimately leading to the expulsion of the foetus (parturition).

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132. (4)

During dehydration, there is a loss of body fluid volume and an increase in ionic concentration in the blood. This activates osmoreceptors in the hypothalamus, which stimulate the release of ADH from the posterior pituitary. ADH promotes water reabsorption in the kidneys, reducing urine output and conserving water. ADH also causes vasoconstriction, which helps maintain blood pressure and glomerular filtration rate (GFR).

(NCERT Ed. 2025-26, 11th, Page No. 212)

133. (1)

List-I	List-II
Gastrin	Acts on gastric glands to secrete HCl and pepsinogen
Secretin	Acts on exocrine pancreas to secrete water and bicarbonate ions
Cholecystokinin (CCK)	Acts on both pancreas and gall bladder to release enzymes and bile
Gastric Inhibitory Peptide	Inhibits gastric secretion and motility

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[Contd...

134. (2)

The given figure represents free central placentation in *Primrose* and *Dianthus*

(NCERT Ed. 2022-23, 11th, Page No. 65)

135. (3)

- The historic Convention on Biological Diversity ('The Earth Summit') was held in Rio de Janeiro.
- The World Summit on Sustainable Development was held in 2002 in Johannesburg, South Africa.

(NCERT Ed. 2025-26, 12th, Page No. 225)

136. (3)

The primary reason for legalizing MTP in India in 1971 was to provide a safe and legal option for women to terminate unwanted pregnancies. This was done to prevent the risks associated with illegal and unsafe abortions performed by unqualified practitioners. While the law also aims to curb female foeticide, the main objective is to ensure the safety and health of women by allowing MTP under strict medical and legal conditions.

(NCERT Ed. 2025-26, 12th, Page No. 46)

137. (1)

NAA → naphthalene acetic acid and 2, 4, D (2, 4-dichlorophenoxy acetic) are synthetic auxins.

(NCERT Ed. 2025-26, 11th, Page No. 175)

138. (2)

The current species extinction rates are estimated to be 100 to 1,000 times faster than in the pre-human times and our activities are responsible for the faster rates.

(NCERT Ed. 2025-26, 12th, Page No. 222)

139. (4)

Sexual reproduction takes place in algae through fusion of two gametes. These gametes can be flagellated and similar in size (as in *Ulothrix*) or non-flagellated (non-motile) but similar in size (as in *Spirogyra*). Such reproduction is called isogamous.

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140. (2)

hCG is produced by the placenta during pregnancy and is responsible for maintaining the corpus luteum, which secretes progesterone to support the pregnancy. hCG is not produced in non-pregnant women.

(NCERT Ed. 2025-26, 12th, Page No. 37)

141. (1)

During translation, the order and sequence of amino acids are defined by the sequence of bases in the mRNA. The amino acids are joined by a bond which is known as a peptide bond.

(NCERT Ed. 2025-26, 12th, Page No. 98)

142. (2)

Non-chordates lack a post-anal tail, which is a defining feature of chordates.

(NCERT Ed. 2025-26, 11th, Page No. 46)

143. (1)

Baculoviruses are pathogens that attack insects and other arthropods.

(NCERT Ed. 2025-26, 12th, Page No. 154)

144. (4)

Common Name	Wheat
Biological Name	*Triticum aestivum*
Genus	*Triticum*
Family	Poaceae
Order	Poales
Class	Monocotyldedonae
Phylum/ Division	Angiospermae

(NCERT Ed. 2022-23, 11th, Page No. 8)

145. (1)

Kelps, which are profusely branched brown algae have a cellulosic wall usually covered on the outside by a gelatinous coating of algin.

(NCERT Ed. 2025-26, 11th, Page No. 27)

146. (3)

Oral contraceptive pills are taken daily for 21 days, followed by a 7-day gap during which menstruation occurs. This cycle is repeated until contraception is no longer desired. Oral contraceptive pills contain small doses of progesterone alone or progesterone estrogen combination.

(NCERT Ed. 2025-26, 12th, Page No. 45)

147. (2)

Sticky ends created by the same restriction enzyme (*Eco*R1) are complementary and can form hydrogen bonds. DNA ligase then stabilizes these bonds by forming phosphodiester linkages.

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[Contd...

148. (3)

List-I	List-II
Gonorrhoea	Completely curable STD
Hepatitis-B	Not completely curable STD
ZIFT	Transfer of the zygote or early embryos (with upto 8 blastomeres) into the fallopian tube
ICSI	A sperm is directly injected into the ovum

(NCERT Ed. 2025-26, 12th, Page No. 48)

149. (1)

- In both lactic acid and alcohol fermentation less than seven per cent of the energy in glucose is released.
- Net gain of ATP in fermentation is 2 ATP.

(NCERT Ed. 2025-26, 11th, Page No. 157)

150. (4)

The medulla oblongata is primarily involved in controlling involuntary functions such as respiration, cardiovascular reflexes, and gastric secretions. It does not control voluntary movements, which are primarily regulated by the cerebral cortex and cerebellum.

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151. (2)

Sickle-cell anaemia is caused by substitution of glutamic acid by valine at sixth position of beta globin chain.

(NCERT Ed. 2025-26, 12th, Page No. 75)

152. (3)

The R group in amino acids determines their classification as acidic, basic, or neutral. For example, glutamic acid has an additional carboxyl group in its R group, which makes it acidic. Lysine is a basic amino acid.

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153. (1)

Kinetin, a modified adenine (purine) was discovered from the autoclaved herring sperm DNA.

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154. (2)

(1) Thomas Hunt Morgan coined the term linkage.
(2) Punnett square used to understand a typical monohybrid cross conducted by Mendel.
(3) Walter Sutton and Theodore Boveri noted that the behaviour of chromosomes was parallel to the behaviour of genes.
(4) Law of segregation was given by Gregor Mendel.

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155. (2)

Taxa can exist at different levels of classification.

(NCERT Ed. 2022-23, 11th, Page No. 5)

156. (2)

The duodenum, the first part of the small intestine, receives bile from the gall bladder and pancreatic juices from the pancreas through a common bile duct.

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157. (4)

The trachea is a rigid tube supported by C-shaped cartilaginous rings that prevent it from collapsing during inhalation and exhalation. It divides into the right and left primary bronchi at the level of the 5th thoracic vertebra. The trachea is lined with ciliated epithelium and mucus-producing cells that trap and remove dust particles. However, the trachea does not collapse during expiration; its cartilaginous rings maintain its patency (openness) at all times. Trachea is part of conducting zone of respiratory system which clears the inhaled air and carry it towards the real exchange part comprises alveoli and their ducts.

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158. (3)

List-I	List-II
Pulmonary circulation	Carries deoxygenated blood from the heart to the lungs and returns oxygenated blood to the heart
Systemic circulation	Supplies oxygenated blood to the entire body and returns deoxygenated blood to the heart
Hepatic portal system	Transports nutrient-rich blood from the digestive organs to the liver for processing
Coronary system	Supplies oxygenated blood to the heart muscle and returns deoxygenated blood to the heart

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[Contd...

159. (2)

The Human Immunodeficiency Virus (HIV), which causes AIDS, primarily infects CD4+ T-helper cells. This leads to their progressive destruction, severely weakening the immune system. As a result, the body becomes vulnerable to opportunistic infections and certain cancers.

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160. (4)

Cyclostomes have a sucking and circular mouth without jaws. Their body lacks scales, unlike Chondrichthyes, which have placoid scales. They have a cartilaginous cranium and vertebral column. They are marine but migrate to freshwater for spawning. They have a closed circulatory system.

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161. (3)

Xylem parenchyma, phloem parenchyma, sieve tubes and companion cells are living at maturity.

(NCERT Ed. 2022-23, 11th, Page No. 87)

162. (3)

List-I	List-II
Hyperglycemic hormone	Glucagon
Hypercalcemic hormone	Parathyroid hormone (PTH)
Gonadotrophic hormone	Luteinizing Hormone (LH)
Blood pressure lowering hormone	Atrial Natriuretic factor (ANF)

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163. (3)

List-I	List-II
Optimum temperature	Temperature for highest enzyme activity
V_{max}	Maximum reaction rate at enzyme saturation
Competitive inhibition	Inhibitor mimics substrate, binds active site
Denaturation	Loss of enzyme structure and function

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164. (3)

The relation between species richness and area for a wide variety of taxa (angiosperm plants, birds, bats, freshwater fishes) turns out to be a rectangular hyperbola. On a logarithmic scale, the relationship is a straight line.

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165. (3)

- In many species of fig trees, there is a tight one-to-one relationship with the pollinator species of wasp.
- It means that a given fig species can be pollinated only by its partner wasp species and no other species.

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166. (4)

The epididymis is located along the posterior surface of the testis, not the anterior surface.

The vasa efferentia transport sperm from the rete testis to the epididymis.

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167. (4)

The given diagram represents spooling process. It is a technique used to isolate and collect DNA from a solution. It involves gently mixing the DNA with ethanol or isopropanol, which causes the DNA to precipitate out of the solution. The precipitated DNA is then "spooled" (wound) onto a glass rod or pipette tip, allowing it to be collected in a visible, thread-like form.



Fig: DNA that separates out can be removed by spooling

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168. (2)

- Chromatids start moving towards opposite poles in anaphase.
- Nuclear envelope develops around the chromosome clusters at each pole forming two daughter nuclei in telophase.

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169. (2)

Sodium ions (Na^+) are essential for generating action potentials in nerve cells. When a nerve is stimulated, Na^+ ions rush into the cell, causing a reversal of polarity across the cell membrane. This reversal is called depolarization and it generates an action potential.

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170. (2)

Insertion of foreign DNA into the tetracycline resistance gene inactivates it, making the transformants sensitive to tetracycline. However, they will still retain resistance to ampicillin, allowing for their selection.

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171. (4)

- Radial vascular bundle : Dicotyledonous root
- Polyarch xylem : Monotyledonous root
- Starch sheath: Dicotyledonous root
- Well developed pith : Monocotyledonous root

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172. (4)

X-rays are ionizing radiations that can cause DNA mutations, leading to cancer, making them carcinogenic.

Benign tumors do not invade surrounding tissues or metastasize to other parts of the body. They remain confined to their site of origin.

Malignant tumors are cancerous and can spread to distant organs through the bloodstream or lymphatic system (metastasis).

Oncogenes are mutated forms of proto-oncogenes that promote uncontrolled cell division, leading to cancer. Tumor suppressor genes (like p53, Rb) normally inhibit cell proliferation and prevent tumor formation.

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173. (3)

Sporopollenin is a complex, resistant organic material that forms the outer layer of the pollen wall (exine).It is resistance to degradation, making it insoluble in acids.

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174. (3)

The correct sequence of electron transport in ETS of thylakoid during non-cyclic photophosphorylation is: PS-II → PQ → Cyt B6f → PC → PS-I

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175. (3)

The equation of *Verhulst-Pearl logistic growth* is $\frac{dN}{dt}=rN\left[\frac{K-N}{K}\right]$. In this equation, K indicates carrying capacity.

(NCERT Ed. 2025-26, 12th, Page No. 195)

176. (3)

Rivets on the wings represent key species.

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177. (4)

Mendelian disorders are mainly determined by alteration or mutation in the single gene. Mendelian disorders are Haemophilia, Cystic fibrosis, Sickle-cell anaemia, Colour blindness, Phenylketonuria, Thalassemia, etc. Turner's syndrome and Down's syndrome are chromosomal disorder.

(NCERT Ed. 2025-26, 12th, Page No. 75)

178. (3)

The number of chromosome and amount of DNA in maize root tip cell in G_1 phase of cell cycle is 40 and 80 pg respectively.

After S phase, the number of chromosomes remain same = 40 and the amount of DNA doubles = 80pg.

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179. (2)

In the grass family the cotyledon is called scutellum that is situated towards one side (lateral) of the embryonal axis.

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180. (1)

Many companies obtain patents for genetic resources that have been traditionally developed and used by local farmers and indigenous communities. This restricts their access to these resources, creating ethical and economic concerns regarding biopiracy and intellectual property rights.

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